

Neural Network Components and Architecture

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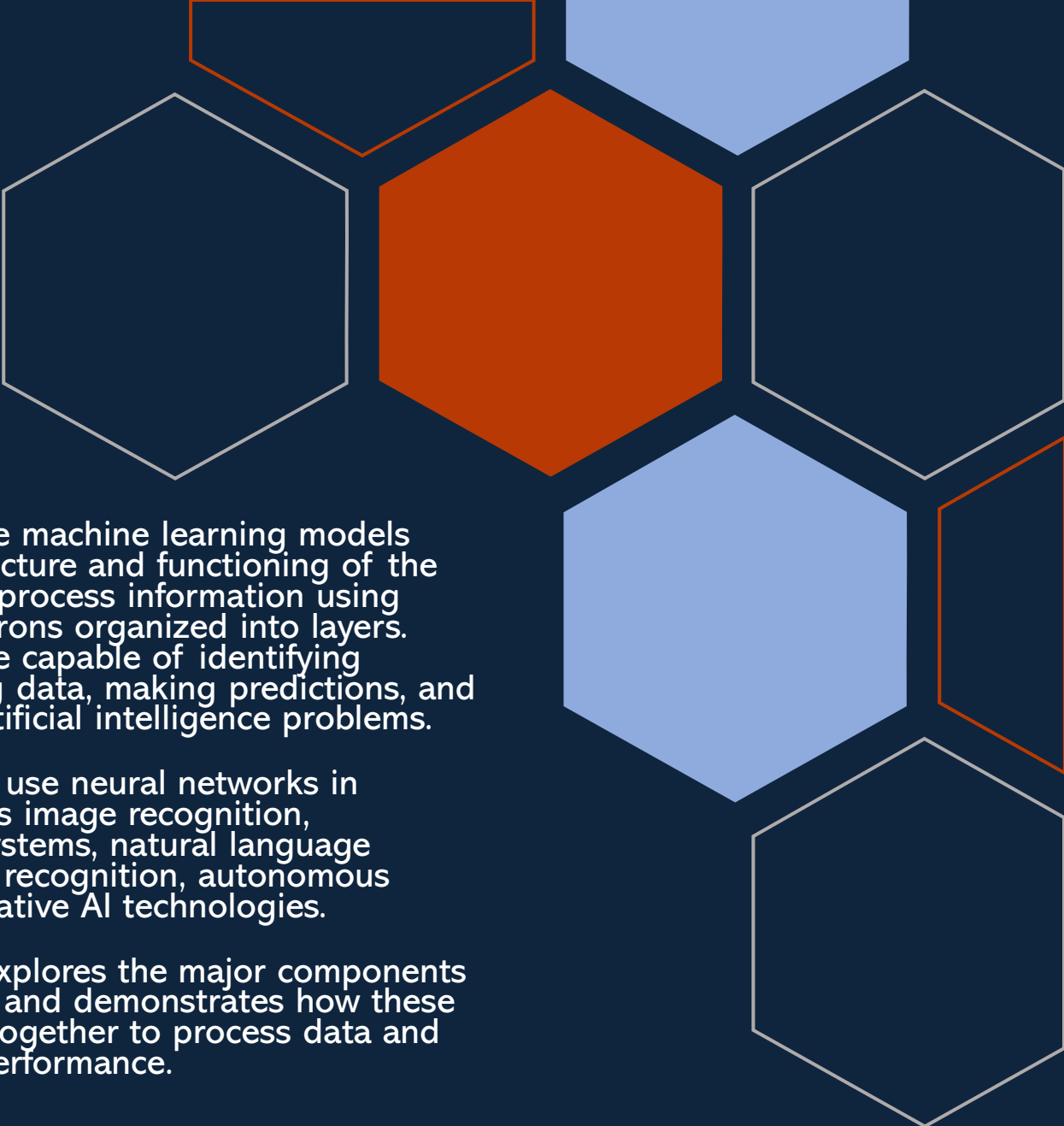
AIML-505 – Large Language and Generative AI





INTRODUCTION TO NEURAL NETWORKS

Introduction to Neural Networks

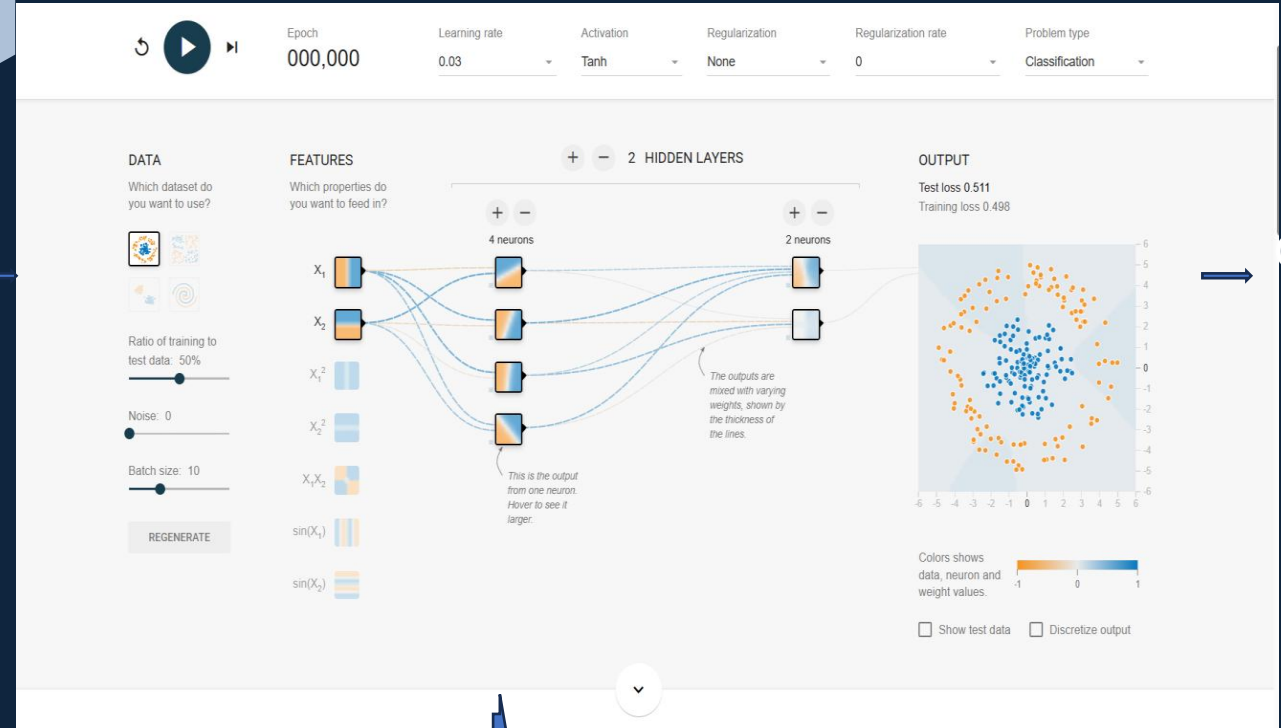
- Neural networks are machine learning models inspired by the structure and functioning of the human brain. They process information using interconnected neurons organized into layers. Neural networks are capable of identifying patterns, classifying data, making predictions, and solving complex artificial intelligence problems.
 - Modern AI systems use neural networks in applications such as image recognition, recommendation systems, natural language processing, speech recognition, autonomous vehicles, and generative AI technologies.
 - This presentation explores the major components of neural networks and demonstrates how these components work together to process data and improve learning performance.
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OVERALL NEURAL NETWORK ARCHITECTURE

Input
Layer

Hidden
Layers

Output
Layer



Layers in Neural Networks

Neural networks are composed of multiple layers that process information step-by-step. Each layer plays an important role in helping the model learn patterns from data.

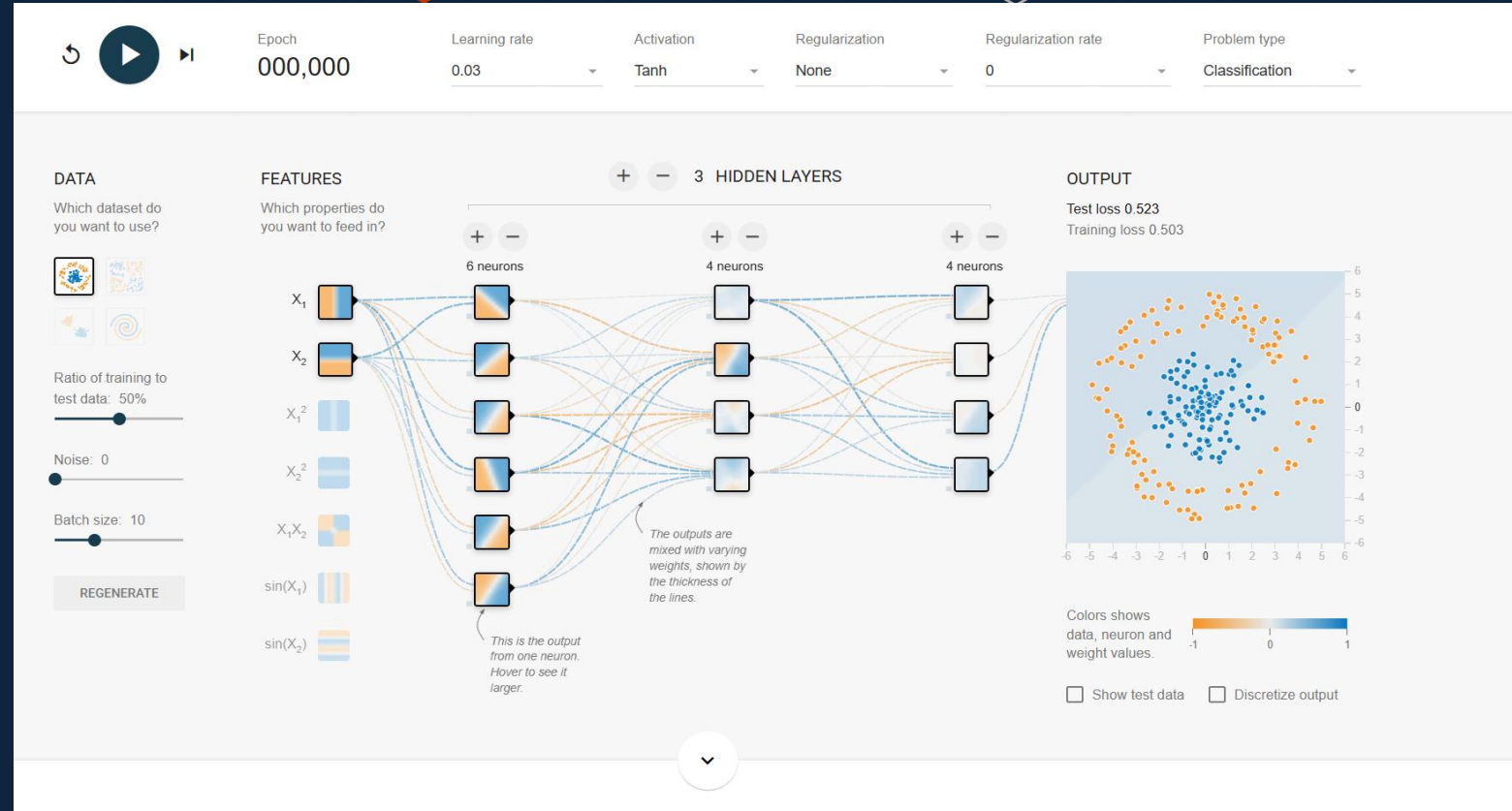
Types of Layers

Input Layer → receives raw data

Hidden Layers → process patterns and relationships

Output Layer → produces final predictions

Hidden layers improve the learning capability of neural networks by identifying complex relationships within datasets.



Additional hidden layers increase network complexity and improve the network's ability to learn patterns from data.

NEURONS AND WEIGHTS

- **Neurons are the fundamental processing units of a neural network. Each neuron receives input values, performs mathematical calculations, and passes information to the next layer.**

Weights determine the strength and importance of connections between neurons. During training, the neural network continuously adjusts weights to improve prediction accuracy and learning performance.

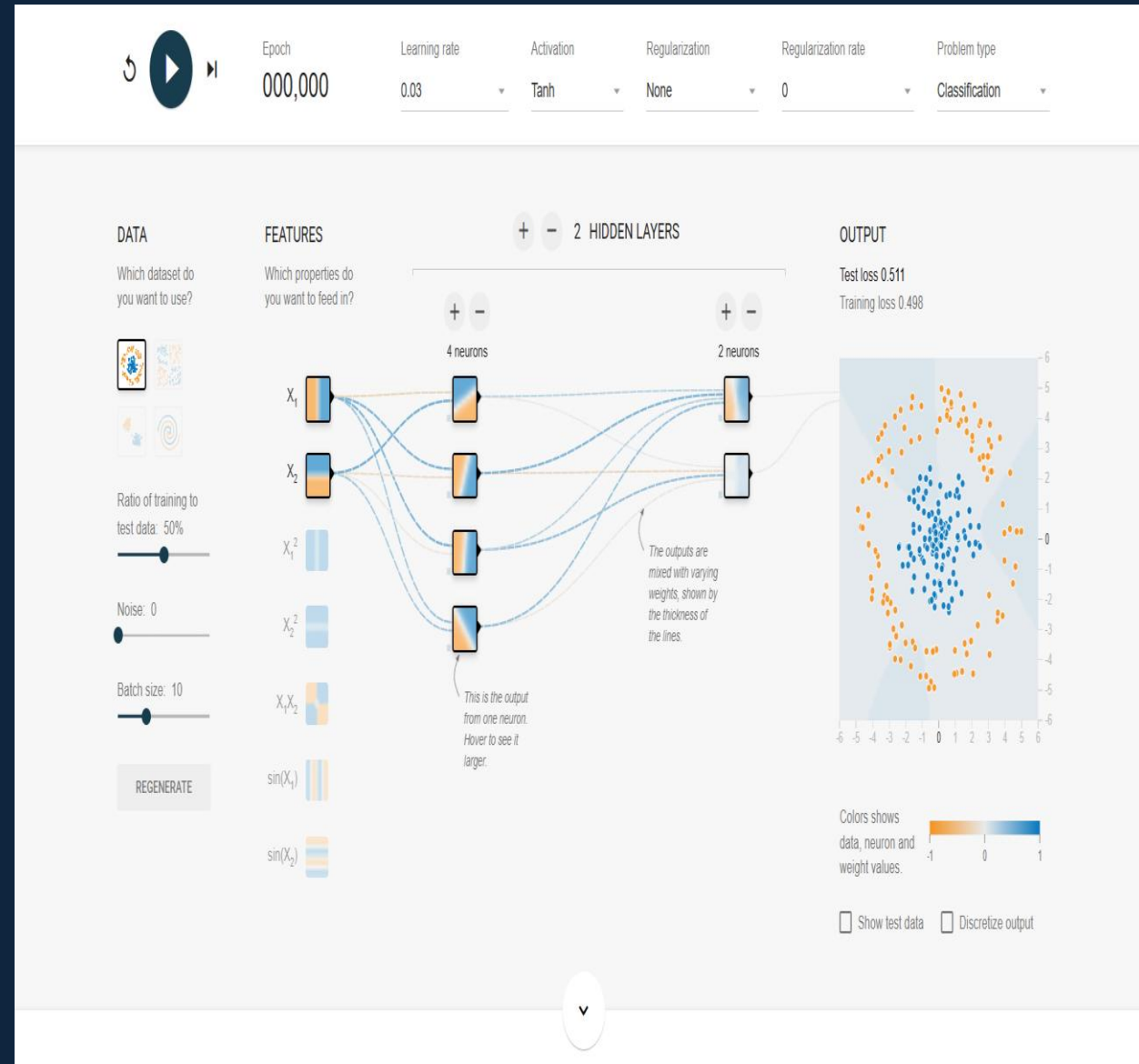
Key Functions

Neurons process input data

Weights influence learning strength

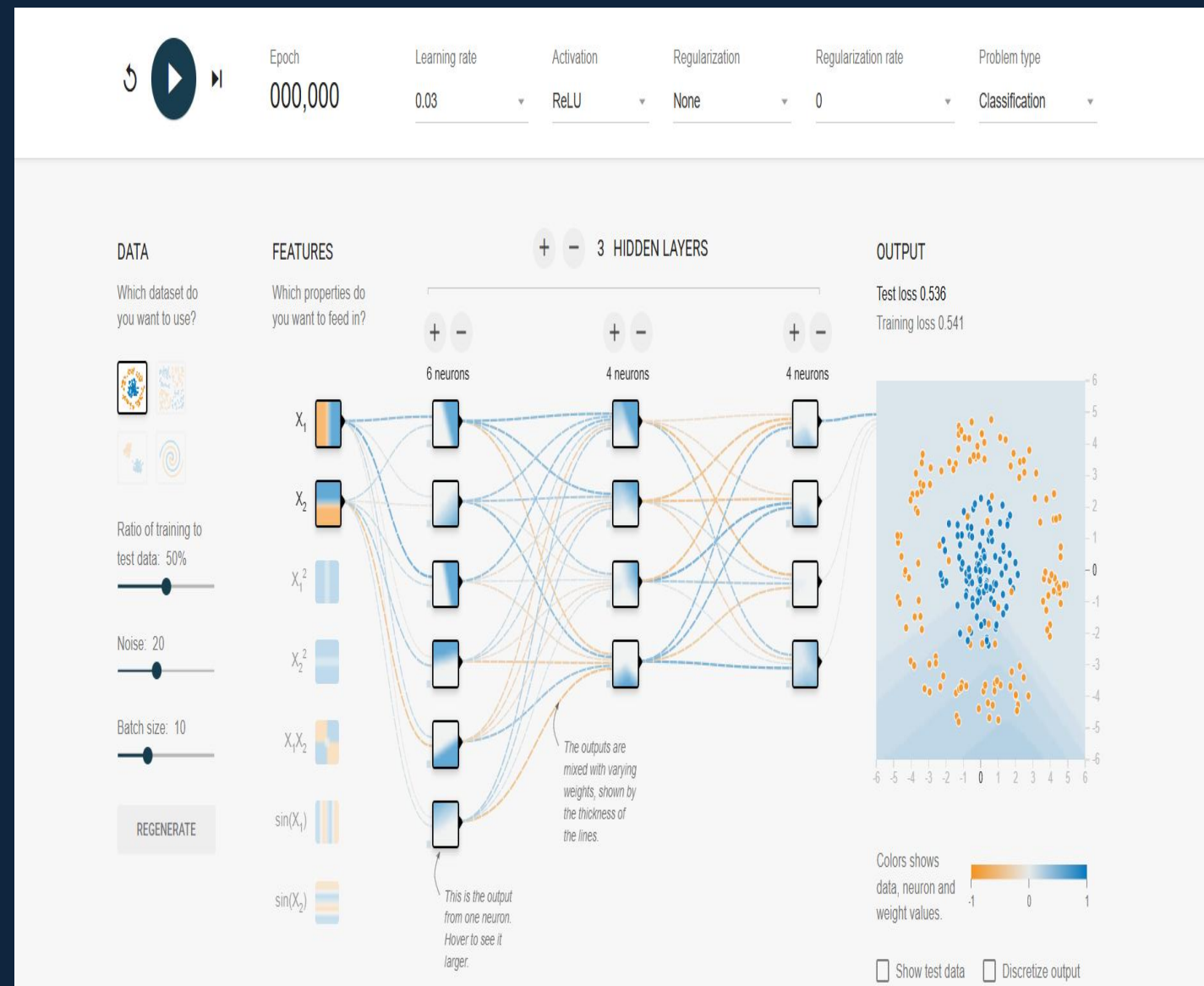
Connections help identify patterns

Weight adjustments improve predictions



Activation Functions

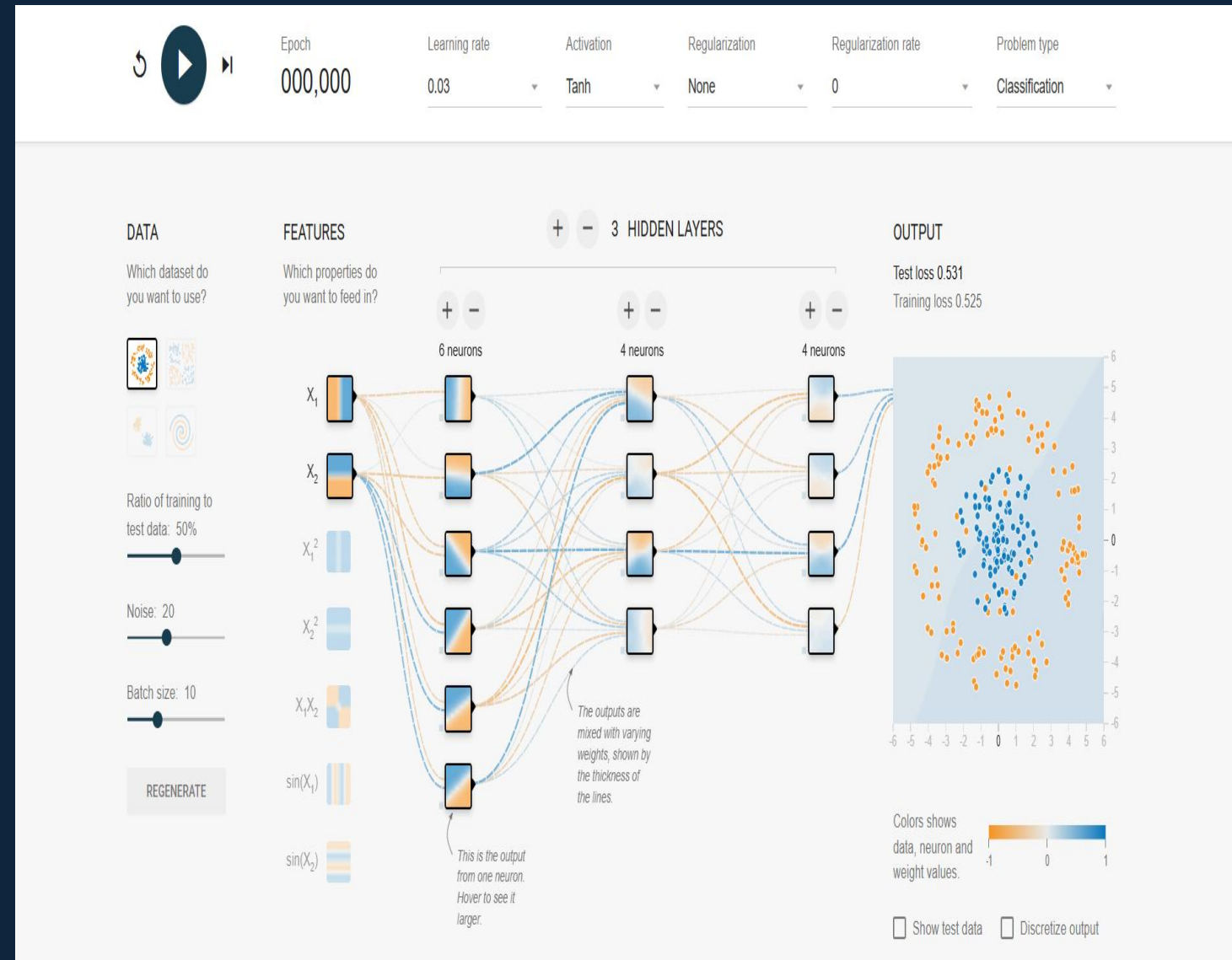
- Activation functions determine whether a neuron should activate and pass information forward through the network. They introduce non-linearity, allowing neural networks to solve complex problems and learn advanced patterns.
- **Common Activation Functions**
- ReLU
- Tanh
- Sigmoid
- Different activation functions impact:
 - training speed
 - prediction accuracy
 - learning performance
- Activation functions play a major role in improving neural network efficiency and optimization.



Changing activation functions alters network learning behavior, prediction patterns, and training performance.

Loss Functions

- Loss functions measure how far the predicted outputs are from the actual expected results during neural network training. They help evaluate model performance and guide the learning process.
- Neural networks continuously minimize loss values to improve prediction accuracy and optimize learning performance.
- **Common Loss Functions**
- Mean Squared Error (MSE)
- Cross Entropy Loss
- Binary Cross Entropy
- **Importance of Loss Functions**
- measure prediction errors
- guide optimization
- improve model accuracy
- support efficient training



Higher noise levels increase prediction difficulty and influence neural network loss during training.

OPTIMIZATION ALGORITHMS

Optimization algorithms improve neural network performance by adjusting weights to minimize loss and improve prediction accuracy during training.

These algorithms help neural networks learn patterns efficiently and converge toward better solutions.

Common Optimization Algorithms

Gradient Descent

Stochastic Gradient Descent (SGD)

Adam Optimizer

Functions of Optimization Algorithms

reduce training loss

improve learning efficiency

update weights automatically

enhance prediction performance

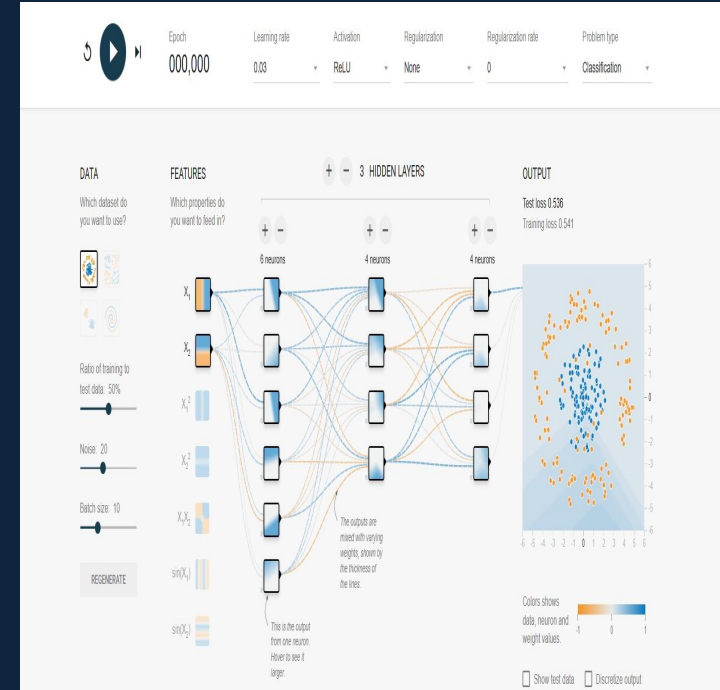
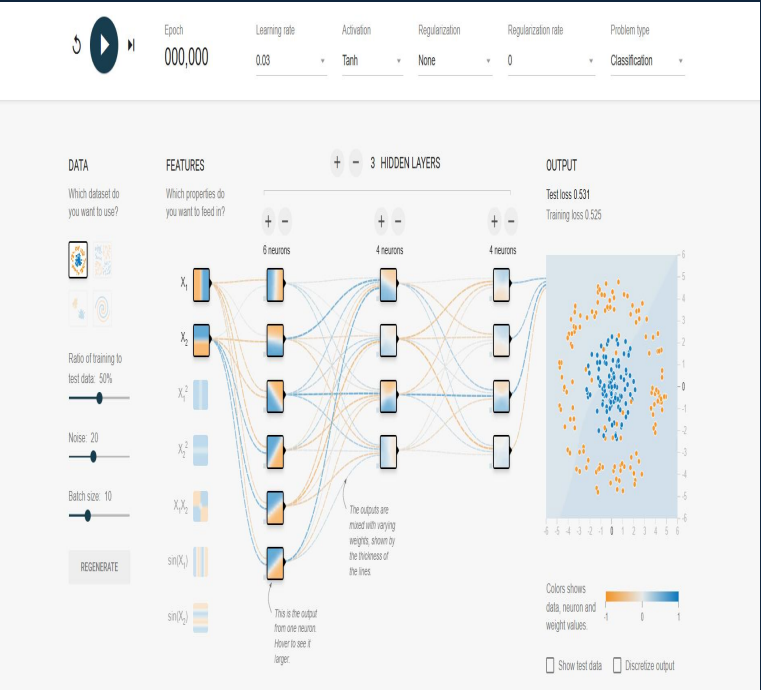
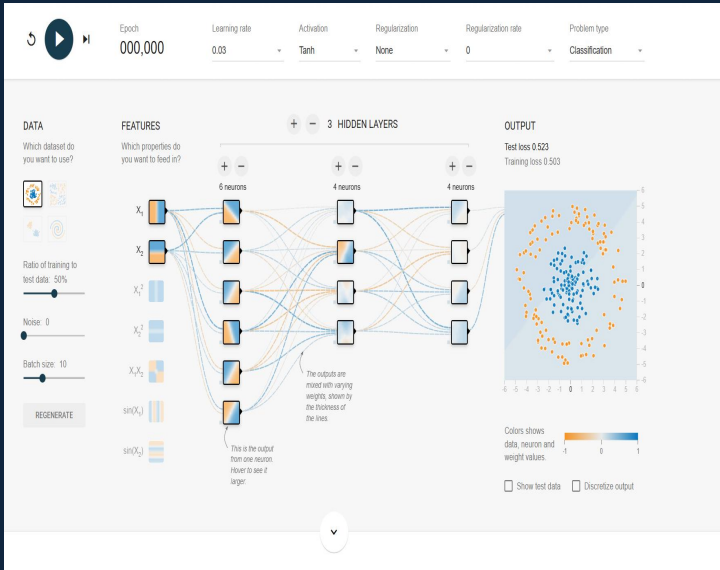
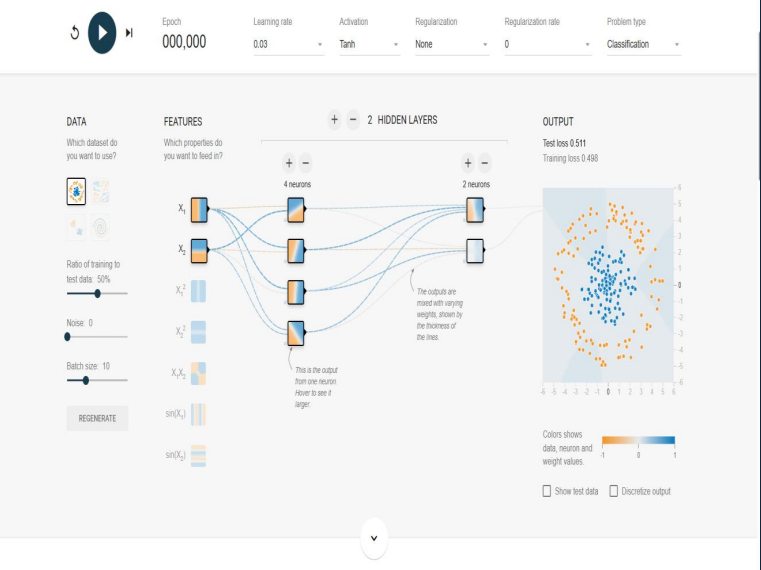
Optimization algorithms are essential for helping neural networks train faster and produce more accurate results.



Increasing noise levels makes training more difficult and impacts optimization performance and prediction accuracy. 8

Neural Network Playground Experiment Results

- The TensorFlow Neural Network Playground was used to explore how different neural network settings impact model performance, learning behavior, and classification accuracy.
- Several experiments were performed by modifying:
- hidden layers
- neuron counts
- activation functions
- noise levels
- training settings
- These experiments provided a practical understanding of how neural networks learn patterns and improve predictions.
- **Key Observations**
- Additional hidden layers increased learning complexity
- Higher noise levels reduced prediction accuracy
- Activation functions changed training behavior
- Network architecture influenced model performance
- Weight adjustments improved classification results
- The playground visually demonstrated how neural networks process data and optimize learning over time.





Summary and Key Insights

- This activity provided a practical understanding of neural network architecture and the major components involved in machine learning systems. The exploration demonstrated how layers, neurons, weights, activation functions, loss functions, and optimization algorithms work together to process information and improve learning performance.
- Using the TensorFlow Neural Network Playground helped visualize:
 - data flow through neural networks
 - relationships between neural network components
 - training behavior and prediction accuracy
 - the impact of hidden layers and activation functions
 - optimization challenges caused by noisy data
- The experiments also demonstrated how neural networks continuously adjust weights and minimize loss to improve classification performance and learning efficiency.
- Overall, this project strengthened understanding of artificial intelligence systems and the practical behavior of neural networks in machine learning applications.



Thank you

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